

QWAI

Your Personal AI Trading Agent for Crypto

Development Milestones & Architecture

6-Week Delivery Timeline

1. Executive Summary

Platform vision and development approach

QWAI is a personal AI trading agent for cryptocurrency markets, delivered as a web platform with an integrated Telegram bot. The system learns individual trading patterns, executes trades autonomously, monitors positions around the clock, and provides proactive intelligence — all through natural language conversation.

This document outlines a 6-week accelerated delivery plan covering the complete product lifecycle from infrastructure setup through production deployment. Each week represents a focused development sprint with clearly defined deliverables and milestone gates.

The architecture follows a modular, event-driven design with clear separation between the intelligence layer (AI agent), execution layer (trade engines), and data layer (market data, analytics). This ensures independent scaling of each subsystem and clean integration points for V2 expansion.

1.1 Project Overview

Parameter	Detail
Timeline	6 Weeks (Accelerated MVP)
Platforms (MVP)	Web Application + Telegram Bot (Mobile App in V2)
Chains Supported	Solana (Jupiter) + EVM Chains (1inch)
AI Engine	LLM-powered conversational agent with persistent memory
Core Differentiator	Personal trading agent that learns, executes, and protects 24/7
Tech Stack	Next.js, NestJS, PostgreSQL, Redis, BullMQ, Docker, AWS

2. System Architecture

High-level architecture overview with layered component breakdown

The QWAI platform follows a layered microservices architecture. Each layer is independently deployable and horizontally scalable. Communication between layers uses a combination of synchronous REST/GraphQL APIs and asynchronous event-driven messaging via BullMQ job queues.

2.1 Architecture Layers

Layer	Components	Description
Client Layer	Next.js Web App Telegram Bot (Grammy)	User-facing interfaces. Web dashboard with embedded chat, TradingView charts, and portfolio UI. Telegram bot for mobile-first commands and notifications.
API Gateway	NestJS API Rate Limiter JWT Auth WebSocket Server	Central API layer handling authentication, request validation, rate limiting, and real-time WebSocket connections for live data streaming.
Intelligence Layer	AI Agent (LLM) Trading DNA Conviction Engine Emotional Intelligence	Core AI brain. Processes natural language, maintains conversational memory, builds user trading profiles, scores tokens, and monitors emotional state.
Execution Layer	Jupiter (Solana) 1inch (EVM) Order Manager DCA Engine Guardrails	Handles all trade execution with smart routing, MEV protection, advanced order types, and mandatory risk guardrails on every transaction.
Data & Analytics	Market Data Service Token Intel On-Chain Analytics News Aggregator	Real-time price feeds, security scanning (GoPlus/RugCheck), whale tracking, sentiment analysis, and news aggregation.
Infrastructure	PostgreSQL Redis BullMQ AWS KMS Docker Datadog	Persistent storage, caching, job queues for background agents, encrypted key management, containerized deployment, and observability.

2.2 Core Data Flow

The following describes the primary data flow for a user trade request through the QWAI system:

- 1. User Input:** User sends a natural language message via Web Chat or Telegram (e.g., "Buy \$200 of SOL"). The message is routed through the API Gateway with JWT authentication.
- 2. AI Processing:** The Intelligence Layer receives the message along with conversation history and Trading DNA profile. The LLM interprets intent, checks emotional state, and formulates a response with a conviction score.
- 3. Risk Validation:** Before any execution, the Guardrails system validates the trade against per-trade limits, daily spend caps, slippage thresholds, and token whitelist/blacklist rules. If the token is flagged by GoPlus or RugCheck, the user is warned.
- 4. Trade Execution:** The Execution Layer routes the order to Jupiter (Solana) or 1inch (EVM) with optimal routing and MEV protection. Advanced order types (stop-loss, trailing stop, bracket) are registered with the background agent framework.
- 5. Confirmation & Monitoring:** Trade confirmation is pushed back to the user in real-time via WebSocket. Background agents begin monitoring the position 24/7 for stop-loss triggers, liquidation risks, and whale movements.
- 6. Learning & Storage:** The trade outcome is logged to the Trading DNA profile. Win/loss patterns, hold times, and behavioral signals are updated for future personalization.

2.3 External Integrations

Service	Purpose	Integration Method
Jupiter Aggregator	Solana DEX Routing	SDK / REST API with Jito bundle submission for MEV protection
1inch	EVM DEX Routing	REST API with gas optimization and split routing
GoPlus Security	EVM Token Safety	REST API for honeypot detection, rug analysis, hidden tax scan
RugCheck	Solana Token Safety	API for mint authority, freeze authority, LP health analysis
CoinGecko / Birdeye	Market Data	REST + WebSocket for real-time price feeds, trending, volume data
TradingView	Charting	Embedded lightweight charts widget in web dashboard
Binance / Bybit / OKX	CEX Integration	User-provided API keys for read-only balance aggregation
Telegram Bot API	Messaging	Grammy framework for bot commands, inline keyboards, push notifications

3. Development Timeline — 6 Weeks

Sprint-by-sprint breakdown with deliverables and milestone gates

Each week represents a focused development sprint. Milestone gates at the end of each week serve as quality checkpoints before advancing to the next phase. All deliverables are cumulative — each week builds on the infrastructure established in prior sprints.

3.1 Timeline Overview

W1	W2	W3	W4	W5	W6
Foundation	Market Data	Trade Execution	Autonomous Agents	Web Platform	Analytics, Social

3.2 Week 1: Foundation & Core Infrastructure

#	Deliverable
1	Project scaffolding: monorepo setup (Turborepo), CI/CD pipeline (GitHub Actions), Docker configs
2	Database schema design and migration setup (PostgreSQL + Prisma ORM)
3	Authentication system: JWT-based auth, wallet connect (Phantom/MetaMask), session management
4	Core wallet engine: encrypted key generation, KMS integration, multi-wallet support (up to 5)
5	Base API gateway with rate limiting, error handling, and request validation
6	WebSocket infrastructure for real-time price feeds and notifications

MILESTONE	Users can sign up, authenticate via wallet, and generate/manage encrypted wallets. Base API and real-time infra operational.
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3.3 Week 2: Market Data & AI Agent Core

#	Deliverable
1	Market data aggregation service: price feeds (CoinGecko/Birdeye), trending tokens, top movers
2	Token intelligence module: GoPlus security scan (EVM), RugCheck (Solana), holder breakdown
3	AI agent core: LLM integration (GPT-4/Claude), prompt engineering pipeline, conversation memory (Redis + Postgres)
4	Trading DNA system: user preference tracking, risk profile builder, behavioral pattern storage

5	Contract address / ticker paste-and-analyze flow: security + sentiment + portfolio fit scoring
6	Conviction scoring engine: multi-signal aggregation into 1–10 score

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Paste any token address and receive a full intelligence card. AI chat responds with personalized, context-aware analysis.

3.4 Week 3: Trade Execution & Risk Engine

#	Deliverable
1	Solana trade execution via Jupiter Aggregator: market buys/sells, smart routing, Jito MEV protection
2	EVM trade execution via 1inch: swaps, routing optimization, gas scheduling
3	Advanced order types: stop-loss, take-profit, trailing stop, bracket orders, limit orders
4	DCA engine: hourly/daily/weekly/monthly recurring buys with configurable parameters
5	Guardrails system: per-trade limits, daily limits, slippage caps, token whitelist, global kill switch
6	Paper trading module: virtual balance tracking, same AI interface, separate P&L ledger

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Full trade execution on Solana and EVM with all order types. Risk guardrails enforced on every transaction. Paper trading functional.

3.5 Week 4: Autonomous Agents & Proactive Intelligence

#	Deliverable
1	Background agent framework: persistent workers for DCAs, stop-losses, copy trades, snipe orders
2	24/7 position monitoring: health factor tracking, liquidation defense triggers, whale movement alerts
3	Proactive notification engine: behavioral triggers (viewed 3x without buying), price alerts, risk flags
4	Morning briefing service: batch-computed daily summary (portfolio delta, executed orders, market overview)
5	Emotional intelligence module: message sentiment analysis, trade frequency tracking, cautionary nudges
6	Agent management dashboard API: view, pause, resume, kill individual or all agents

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QWAI operates autonomously 24/7. Proactive alerts, morning briefings, and emotional intelligence guardrails active.

3.6 Week 5: Web Platform & Telegram Bot

#	Deliverable
1	Web dashboard: full-screen charts (TradingView), portfolio breakdown, trade journal, analytics panels
2	Embedded QWAI chat interface on dashboard with streaming responses
3	Wallet management UI: deposit, withdraw, export keys, multi-wallet switching
4	Telegram bot: natural language chat, quick trade commands, push notifications, inline buttons
5	Unified session layer: same AI memory across Web and Telegram
6	Notification preferences: Telegram, email, Discord webhook configuration

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Complete web platform and Telegram bot live. Users interact with same AI brain across both interfaces.

3.7 Week 6: Analytics, Social & Production Hardening

#	Deliverable
1	Performance analytics: trade replay, win rate, avg hold time, risk-adjusted returns, weekly review
2	On-chain analytics: wallet analysis, holder distribution, whale tracking, correlation analysis
3	Copy trading engine: mirror verified wallet trades in real-time
4	Social features: trading rooms, leaderboard (anonymized), signal sharing cards, referral system
5	Exchange integration: Binance, Bybit, OKX API key connection for unified portfolio view
6	Production hardening: load testing, security audit, monitoring (Datadog), logging, staging → production deploy

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Full product launch-ready. Analytics, social features, exchange integrations, and copy trading operational. Production deployed.

4. Feature Module Map

Core product capabilities organized by functional domain

Module	Key Capabilities	Delivery Week
AI Agent	Natural language chat, trade execution via conversation, conviction scoring, personalized responses	Week 2
Trading DNA	Behavioral profiling, pattern recognition, preference learning, session memory	Week 2
Trade Execution	Jupiter (SOL), 1inch (EVM), all order types, DCA, MEV protection, multi-wallet buy	Week 3
Risk Engine	Guardrails, kill switch, GoPlus/RugCheck scans, portfolio concentration warnings, risk simulator	Week 3
Autonomous Agents	24/7 monitoring, background workers, liquidation defense, whale alerts, copy trading	Week 4
Proactive Intel	Morning briefings, emotional intelligence, trigger-based notifications, gas scheduling	Week 4
Web Platform	Dashboard, charts, portfolio view, trade journal, embedded chat, wallet management	Week 5
Telegram Bot	Chat interface, quick trades, push notifications, inline actions, same AI brain	Week 5
Analytics	Trade replay, performance metrics, on-chain analysis, backtesting, tax reporting	Week 6
Social Layer	Trading rooms, leaderboard, signal sharing, copy trading, referral system	Week 6

5. Technology Stack

Core technologies powering the QWAI platform

Category	Technology	Purpose
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Frontend	Next.js 14, TailwindCSS, TradingView	Web dashboard, responsive UI, real-time charts
Backend	NestJS, Prisma ORM, BullMQ	API services, database access, background job queues
Database	PostgreSQL, Redis	Persistent storage, caching, session/memory store
AI / LLM	GPT-4 / Claude API, LangChain	Conversational agent, intent parsing, analysis
Blockchain	Jupiter SDK, 1inch API, Solana Web3.js, Ethers.js	Trade execution, wallet management, on-chain reads
Security	AWS KMS, GoPlus API, RugCheck API	Key encryption, token safety scanning
Messaging	Grammy (Telegram), WebSocket	Bot framework, real-time push notifications
DevOps	Docker, GitHub Actions, AWS ECS, Datadog	Containerization, CI/CD, hosting, monitoring

6. Risk Matrix & Mitigation

Key risks identified for the 6-week delivery timeline

Risk	Impact	Likelihood	Mitigation Strategy
DEX API instability	High	Medium	Implement fallback routing, circuit breakers, and retry logic. Maintain secondary aggregator integrations.
LLM latency spikes	Medium	Medium	Use streaming responses, implement response caching for common queries, set hard timeout limits with graceful fallback.
Key security breach	Critical	Low	AWS KMS for all key encryption. No plaintext keys in memory. Regular security audits and penetration testing.
Scope creep	High	High	Strict weekly milestone gates. Features not in the current sprint are deferred. Product owner sign-off required for scope changes.
Third-party rate limits	Medium	High	Implement request pooling, caching layers, and graceful degradation. Monitor quota usage with alerts at 80% threshold.
Telegram bot downtime	Medium	Low	Redundant bot instances, health checks, auto-restart policies. Web platform serves as fallback interface.

7. Acceptance Criteria

Definition of done for final delivery

1. User can sign up, connect wallet, and generate encrypted wallets with full self-custody
2. AI chat agent responds with personalized, context-aware trading analysis and executes trades on command
3. Paste any token address and receive a complete intelligence card with security scan, holder data, and conviction score
4. All trade types functional: market, limit, stop-loss, take-profit, trailing stop, DCA, bracket orders
5. Guardrails enforced on every trade: spend limits, slippage caps, whitelist, kill switch operational
6. Paper trading mode available with separate P&L tracking and one-command switch to live trading

7. Background agents run 24/7: position monitoring, DCA execution, copy trades, liquidation defense
8. Morning briefing delivered daily via Telegram push notification
9. Emotional intelligence nudges active during high-frequency or late-night trading sessions
10. Web dashboard fully functional with charts, portfolio view, trade journal, and embedded chat
11. Telegram bot operational with natural language commands, inline actions, and push notifications
12. Performance analytics: trade replay, win rate, behavioral insights, weekly review
13. Exchange API integration for unified portfolio view across Binance, Bybit, and OKX
14. Production deployed with monitoring, logging, and security hardening complete

8. V2 Roadmap — Post-MVP

Features and expansions planned for the second major release

The following capabilities are intentionally scoped out of the 6-week MVP delivery to maintain focus and timeline integrity. They represent high-value enhancements that build on the foundation established in V1.

8.1 Dedicated Mobile Application

A native mobile app (React Native) will serve as the primary V2 deliverable, bringing the full QWAI experience to iOS and Android with platform-specific optimizations:

- Home screen widgets displaying real-time portfolio value, active agents, and top movers
- Push notifications with quick-action buttons for trade confirmations and alert responses
- Biometric authentication (Face ID / fingerprint) for trade confirmations and wallet access
- Offline mode with cached portfolio data and queued commands that execute on reconnection
- Native charting experience optimized for touch interactions and mobile screen sizes
- Deep linking from Telegram notifications directly into relevant app screens

8.2 Additional V2 Features

Feature	Description	Priority
Mobile App (iOS/Android)	Full-featured React Native app with widgets, biometric auth, and native push notifications	P0 — Critical
Multi-Language Support	Localized AI responses and UI for key markets: Korean, Japanese, Chinese, Spanish	P1 — High
Advanced Backtesting	Historical strategy simulation with custom parameters, drawdown analysis, and Monte Carlo modeling	P1 — High
Institutional API	White-label API access for funds and trading desks with custom rate limits and dedicated support	P2 — Medium
Cross-Chain Bridges	Native bridge integrations for seamless asset movement between Solana, Ethereum, Arbitrum, and Base	P2 — Medium

Social Trading V2	Verified trader profiles, strategy marketplace, and revenue sharing for signal providers	P2 — Medium
Options & Perps	Perpetual futures and options trading integration via GMX, Drift, and Hyperliquid	P3 — Future